Complete Proofs of Lemma 2 and Lemma 4

Lemma 2. *The maximum deviation between the* PTSoftplus *and the* Softplus *function is bounded by 0.914, i.e.*

$$\|\mathrm{PTSoftplus} - \mathrm{Softplus}\|_\infty \le 0.914.$$

Moreover, the maximum deviation between the derivatives of PTSoftplus *and* Softplus *is bounded by 0.371, i.e.*

$$\|\mathrm{PTSoftplus}' - \mathrm{Softplus}'\|_\infty \le 0.371.$$

Proof. Define $f(x) = \mathrm{PTSoftplus}(x) - \mathrm{Softplus}(x)$. To find an upper bound on the maximum deviation between PTSoftplus and Softplus, it suffices to determine the minimum and maximum values of the function f:

$$f(x) = \begin{cases} 2^x - \ln(1+e^x), & \text{if } x < x_c, \\ x - \ln(1+e^x) + \frac{1}{\ln 2} - \log_2(\frac{1}{\ln 2}), & \text{otherwise.} \end{cases}$$

As $x \to -\infty$, the value of $f(x)$ approaches 0, and as $x \to \infty$, the value of $f(x)$ approaches the constant value $\frac{1}{\ln 2} - \log_2(\frac{1}{\ln 2})$. From Lemma 1, we know that $\mathrm{PTSoftplus}'$ is continuous, and since $\mathrm{Softplus}'$ is also continuous, we know that f is continuously differentiable. The derivative of f is

$$f'(x) = \begin{cases} (\ln 2) \cdot 2^x - \frac{e^x}{1+e^x}, & \text{if } x < x_c, \\ \frac{1}{1+e^x}, & \text{otherwise.} \end{cases}$$

It is clear that, for $x \ge x_c$, the derivative $f'(x)$ is strictly positive, so $f(x)$ is monotonically increasing when $x \ge x_c$. Next, we show that $(\ln 2) \cdot 2^x > \frac{e^x}{1+e^x}$ when $x < x_c$, and consequently $f(x)$ is monotonic on $\mathbb{R}$. By the fact that $2^x = e^{x \ln 2}$, the inequality can be reformulated as

$$(\ln 2) \cdot e^{x \ln 2} > \frac{e^x}{1+e^x}. \tag{28}$$

Divide both sides by $e^{x \ln 2}$. Then the above becomes

$$\ln 2 > \frac{e^{x(1-\ln 2)}}{1+e^x}.$$

Define $g(x)$ as the function on the right-hand side, where

$$g'(x) = \frac{e^{x(1-\ln 2)} \cdot (1 - \ln 2 - (\ln 2) \cdot e^x)}{(1+e^x)^2}.$$

Since $(1+e^x)^2$ and $e^{x(1-\ln 2)}$ are always positive, setting the first derivative $g'(x)$ to 0 gives

$$1 - \ln 2 - (\ln 2) \cdot e^x = 0 \quad \Rightarrow \quad x = \ln\Big(\frac{1-\ln 2}{\ln 2}\Big).$$

Denote the critical point $x_g = \ln(\frac{1-\ln 2}{\ln 2})$. It can be easily checked that $g'(x) > 0$ when $x < x_g$, and $g'(x) < 0$ when $x > x_g$. Therefore, g is a concave function whose maximum value is

$$g(x_g) = \frac{e^{x_g(1-\ln 2)}}{1+e^{x_g}} \le 0.540 < \ln 2 \ (\approx 0.693),$$

which validates the inequality (28). Consequently, $f'(x)$ is strictly positive on $\mathbb{R}$, and f is a monotonic function. Since $f(x) \to 0$ as $x \to -\infty$ and $f(x) \to \frac{1}{\ln 2} - \log_2(\frac{1}{\ln 2})$ as $x \to \infty$, we conclude that

$$\|f\|_\infty = \sup_{x\in\mathbb{R}} |f(x)| \le \frac{1}{\ln 2} - \log_2\big(\frac{1}{\ln 2}\big) \le 0.914.$$

Next, we show that the difference between the derivatives of PTSoftplus and Softplus is also bounded – that is, the supremum norm of f' is bounded. From the above analyses, we already know that $f'(x)$ is always positive on $\mathbb{R}$. It remains to find an upper bound on the value of $f'(x)$.

It is clear that for $x \ge x_c$, $f'(x)$ is strictly decreasing. Therefore, on $[x_c, \infty)$, the maximum value of $f'(x)$ is

$$f'(x_c) = \frac{1}{1+e^{x_c}} \le 0.371.$$

For $x < x_c$, the second-order derivative of f is

$$f''(x) = (\ln 2)^2 \cdot 2^x - \frac{e^x}{(1+e^x)^2}.$$

In what follows, we show that the above $f''(x) > 0$ on $(-\infty, x_c)$, thus $f'(x)$ is strictly increasing. In other words, we want to show

$$(\ln 2)^2 > \frac{e^{x(1-\ln 2)}}{(1+e^x)^2}. \tag{29}$$

Define $h(x)$ to be the function on the right-hand side, whose derivative is

$$h'(x) = \frac{e^{x(1-\ln 2)} \cdot (1+e^x) \cdot [(1-\ln 2)(1+e^x) - 2e^x]}{(1+e^x)^4}.$$

Since $e^{x(1-\ln 2)} \cdot (1+e^x)$ is strictly positive, setting $h'(x)$ to zero gives

$$(1-\ln 2)(1+e^x) - 2e^x = 0 \quad \Rightarrow \quad x = \ln\Big(\frac{1-\ln 2}{1+\ln 2}\Big).$$

Denote the critical point $x_h = \ln(\frac{1-\ln 2}{1+\ln 2})$. It can be easily checked that $h'(x) > 0$ when $x < x_h$, and $h'(x) < 0$ when $x > x_h$. Therefore, h is a concave function whose maximum value is

$$h(x_h) = \frac{e^{x_h(1-\ln 2)}}{(1+e^{x_h})^2} \approx 0.424 \le (\ln 2)^2 \ (\approx 0.480),$$

which validates the inequality (29). Consequently, $f''(x)$ is strictly positive and $f'(x)$ is increasing on $(-\infty, x_c)$. By the fact that $f'(x)$ is continuous, we conclude that

$$\|f'\|_\infty = \sup_{x\in\mathbb{R}} |f'(x)| = f'(x_c) \le 0.371,$$

which completes the proof. $\square$

Lemma 4. *The maximum deviation between the* PTSiLU *and the* SiLU *function is bounded by 0.316, i.e.*

$$\|\text{PTSiLU} - \text{SiLU}\|_\infty \leq 0.316.$$

Moreover, the maximum deviation between the derivatives of PTSiLU *and* SiLU *is bounded by* 0.263*, i.e.*

$$\|\text{PTSiLU}' - \text{SiLU}'\|_\infty \leq 0.263.$$

Proof. Define $\bar{f}(x) = \text{PTSiLU}(x) - \text{SiLU}(x)$. Then the analysis of the deviation between PTSiLU and SiLU reduces to the analysis of $\bar{f}$. We start from the derivatives of $\bar{f}$. The first derivative of $\bar{f}$ is

$$\bar{f}'(x) = \begin{cases} -(\ln 2) \cdot 2^x - \frac{e^x \cdot (1+x+e^x)}{(1+e^x)^2}, & \text{if } x < \bar{x}_c, \\ 1 - (\ln 2) \cdot 2^{-x-1} - \frac{e^x \cdot (1+x+e^x)}{(1+e^x)^2}, & \text{otherwise.} \end{cases}$$

On $(-\infty, \bar{x}_c)$, the second derivative of $\bar{f}$ is

$$\bar{f}''(x) = -(\ln 2)^2 \cdot 2^x - \frac{e^x(2 + x + e^x(2-x))}{(1+e^x)^3},$$

and on $[\bar{x}_c, \infty)$, the second derivative of $\bar{f}$ is

$$\bar{f}''(x) = (\ln 2)^2 \cdot 2^{-x-1} - \frac{e^x(2 + x + e^x(2-x))}{(1+e^x)^3}.$$

Denote $\bar{h}''(x)$ as the function $\frac{e^x(2+x+e^x(2-x))}{(1+e^x)^3}$, which can be simplified to

$$\bar{h}''(x) = \underbrace{\frac{2e^x}{(1+e^x)^2}}_{A(x)} + \underbrace{\frac{x(e^x - e^{2x})}{(1+e^x)^3}}_{B(x)}. \tag{30}$$

Here, the first term $A(x)$ is always positive. For the second term $B(x)$, since $(1+e^x)^3 > 0$, its sign is determined by the numerator $x(e^x - e^{2x})$. When $x < 0$, we have $(e^x - e^{2x}) > 0$, so $B(x) < 0$; when $x > 0$, we have $(e^x - e^{2x}) < 0$, so $B(x)$ is also negative. Therefore, $B(x)$ has a maximum value of 0 at $x = 0$. Since $A(x) > 0$ and $B(x) \leq 0$, to bound the values of $\bar{h}''(x)$, we can find extrema of $A(x)$ and $B(x)$ separately.

The derivative of $A'(x)$ is

$$A'(x) = \frac{2(e^x - e^{2x})}{(1+e^x)^3}.$$

Setting it to zero gives $x = 0$. Moreover, $A'(x) > 0$ when $x < 0$, and $A'(x) < 0$ when $x > 0$. Therefore, $A(x)$ attains its maximum value at $x = 0$, where $A(0) = 0.5$. For the second term $B(x)$, it can be checked that $B(x)$ is symmetric (i.e. $B(x) = B(-x)$). Moreover, by the fact that $0 < \frac{1-e^x}{(1+e^x)^2} < 1$ for $x < 0$, we have

$$B(x) = \frac{xe^x(1-e^x)}{(1+e^x)^3} \geq xe^x \quad \text{for } x < 0.$$

The function $-xe^x$ attains its minimum value of $-\frac{1}{e}$ at $x = -1$. Therefore, we have $B(x) \geq -\frac{1}{e} \geq -0.368$. Together with (30), we have $\|\bar{h}''\|_\infty \leq \max\{\sup|A(x)|, \sup|B(x)|\} = 0.5$. Therefore, for $\bar{f}''$, since $-(\ln 2)^2 \cdot 2^x$ is strictly decreasing on $(-\infty, \bar{x}_c)$, we have

$$\sup_{x \in (-\infty, \bar{x}_c)} |\bar{f}''(x)| \leq (\ln 2)^2 \cdot 2^{\bar{x}_c} + \|\bar{h}''\|_\infty \leq 0.639.$$

On the other hand, since $(\ln 2)^2 \cdot 2^{-x-1}$ is strictly decreasing on $[\bar{x}_c, \infty)$, we have

$$\sup_{x \in [\bar{x}_c, \infty)} |\bar{f}''(x)| \leq (\ln 2)^2 \cdot 2^{-\bar{x}_c - 1} + \|\bar{h}''\|_\infty \leq 1.332.$$

Combining the above two yields $\|\bar{f}''\|_\infty \leq 1.332$. In other words, the rate of change of $\bar{f}'(x)$ is bounded by 1.332.

Based on the above bound on $\|\bar{f}''\|_\infty$, we employ a computer aided analysis for $\bar{f}'$ and $\bar{f}$. We generate a sequence of points $\mathcal{X} = \{x_1, x_2, \ldots, x_N\}$ from -10 to 10, where $x_{i+1} - x_i = 0.001$ for $i = 1, \ldots, N-1$. Then we compute the value of $\bar{f}'(x)$ for each $x \in \mathcal{X}$. The numerical results show that these values are between -0.178 and 0.261. By the fact that $\|\bar{f}''\| \leq 1.332$, we have

$$\sup_{x \in [-10, 10]} |\bar{f}'(x)| \leq 0.261 + 1.332 \cdot 0.001 \leq 0.263.$$

It can be shown that $\bar{f}'(x)$ diminishes when $x \notin [-10, 10]$.

Next, we evaluate the value of $\bar{f}(x)$ for each $x \in \mathcal{X}$. The numerical results show that these values are between -0.229 and 0.315. By the fact that $\|\bar{f}'\| \leq 0.263$, we have

$$\sup_{x \in [-10, 10]} |\bar{f}(x)| \leq 0.315 + 0.263 \cdot 0.001 \leq 0.316.$$

It is easily seen that $\bar{f}(x)$ is negligible when $x < -10$. For $x > 10$, the function value $\bar{f}(x)$ eventually decreases to the limit $\lim_{x \to \infty} \bar{f}(x) \geq -0.229$. $\square$